SMART INDIA HACKATHON 2025

- **Problem Statement ID - SIH25213.**
- **Problem Statement Title - Integrated Wearable Device for Real-Time Monitoring of Vital Parameters, Gas Exposure, and Fatigue.**
- **Theme - Disaster Management.**
- **PS Category - Hardware**
- **Team ID -106308**
- **Team Name - InnovHers.**

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IDEA TITLE

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Problem Statement:

- Rescuers face toxic gas and health risks.
- Early alerts save lives.
- Wireless data assists remote teams.
- Lightweight design ensures comfort.

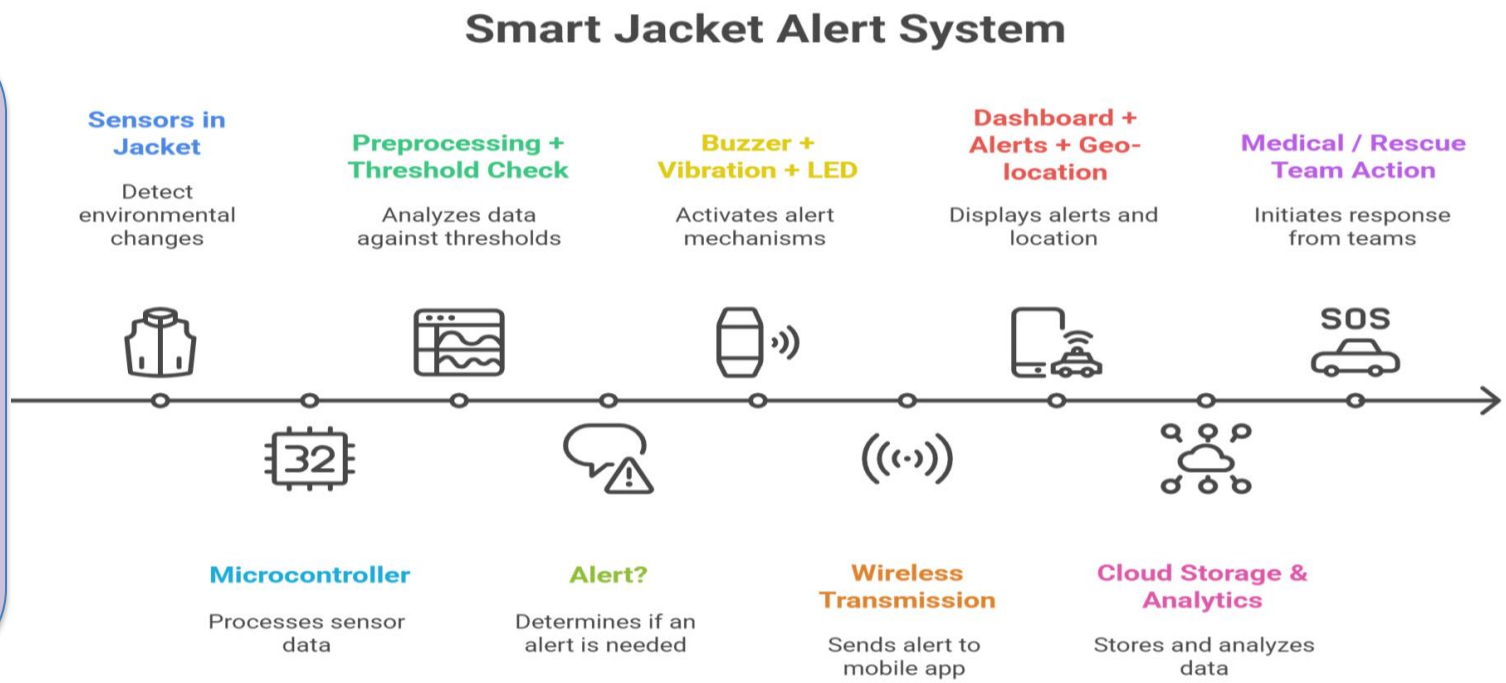
Proposed Solution:

A smart life jacket that uniquely monitors both vital signs and toxic gases in real time for disaster responders.

- Real-time vitals & gas monitoring
- Early risk detection & alerts
- Rescuers & medical teams with live data
- Safety, intervention speed & efficiency

Target Users:

- Rescures
- FireFighters
- Industrial workers
- Emergency medical staff



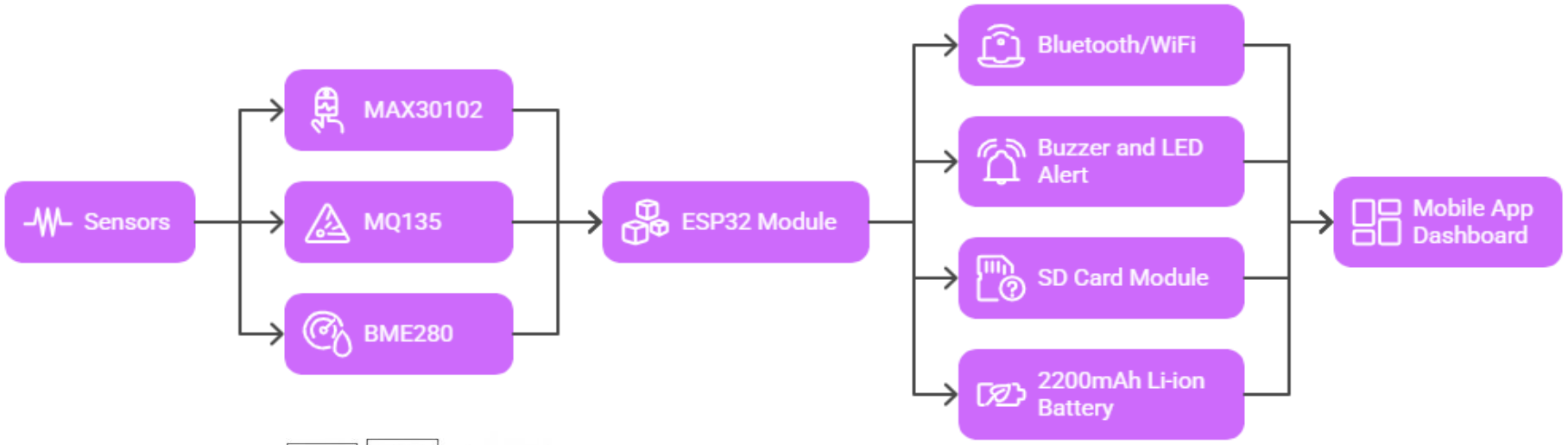


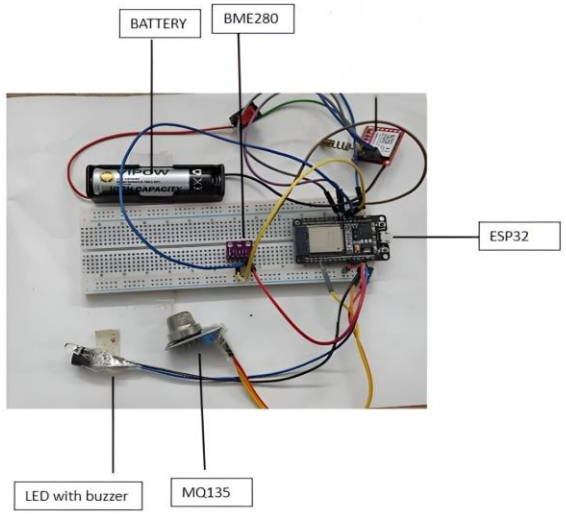
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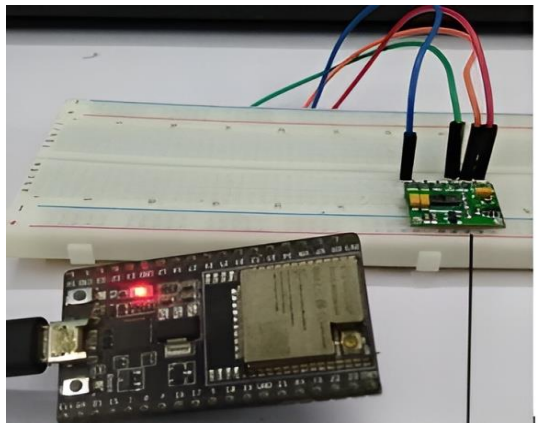
TECHNICAL APPROACH

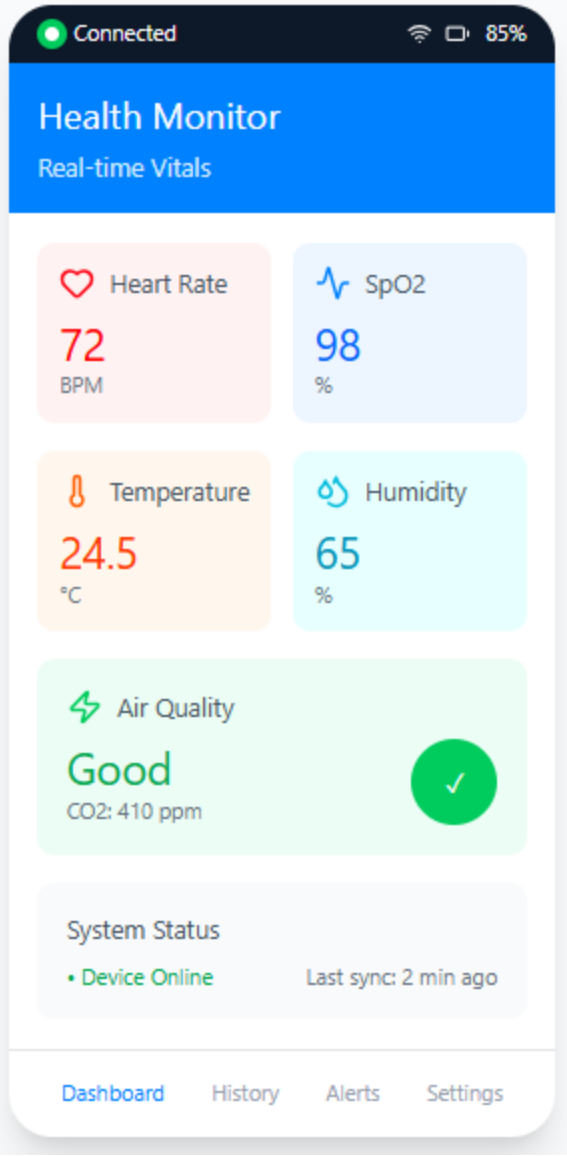
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Technical Approach in Prototype









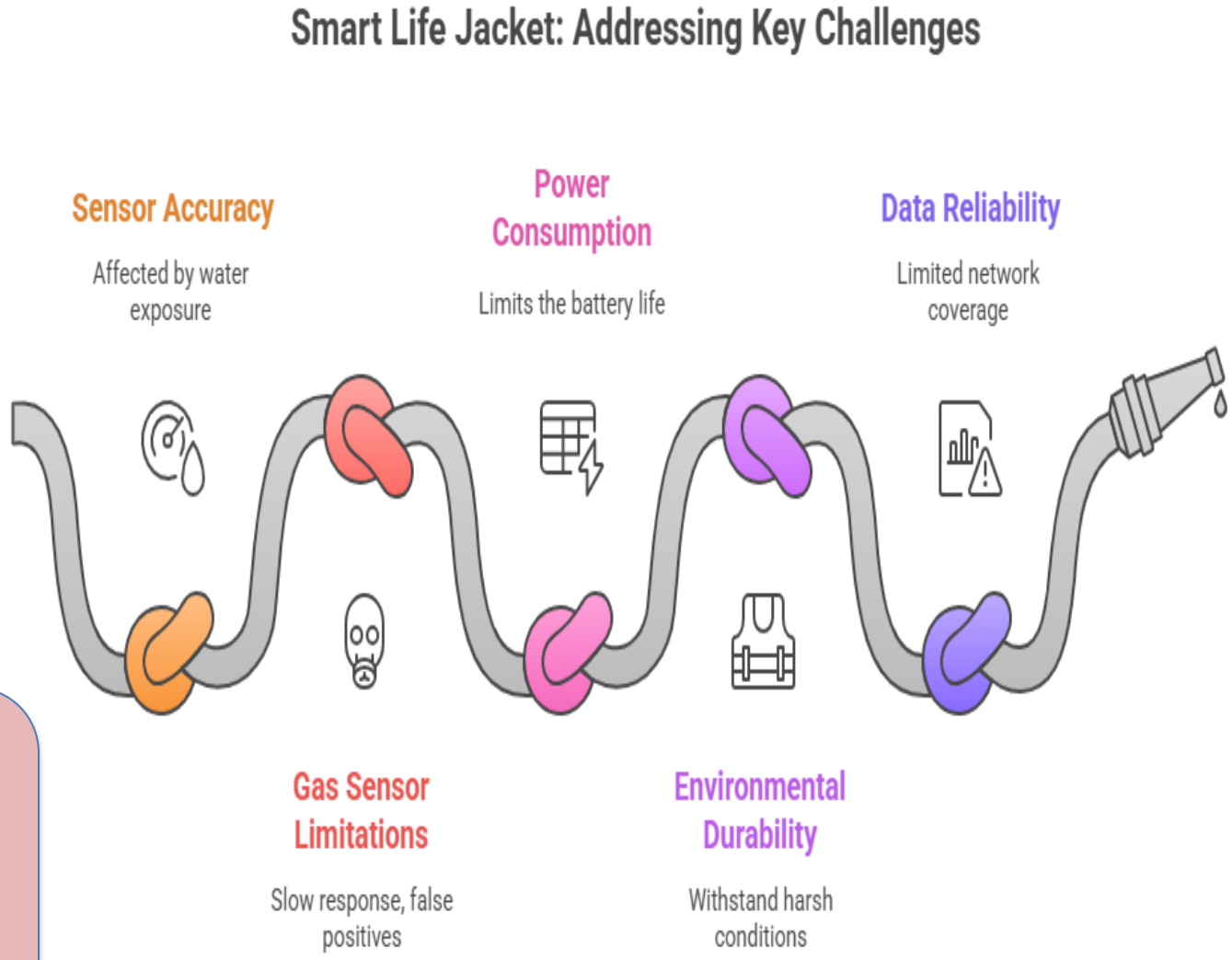
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FEASIBILITY AND VIABILITY

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Strategies for industrial level:

Feature	Hardware	Software
Components	MAX86150, CO, O_2, NH_3, VOCs	Sensor fusion, data filtering
MCU	STM32 Industrial MCU	Bluetooth, Wi-Fi, LoRaWAN protocol
Connectivity	LoRaWAN + Bluetooth backup	Android, Azure IoT
Power	Swappable Li-ion 5000mAh	Web-based monitoring console
Safety	Flame-retardant (Nomex)	Predictive health risk analysis
Alerts	LED, buzzer	Fatigue/exposure detection



Challenges and Risks:

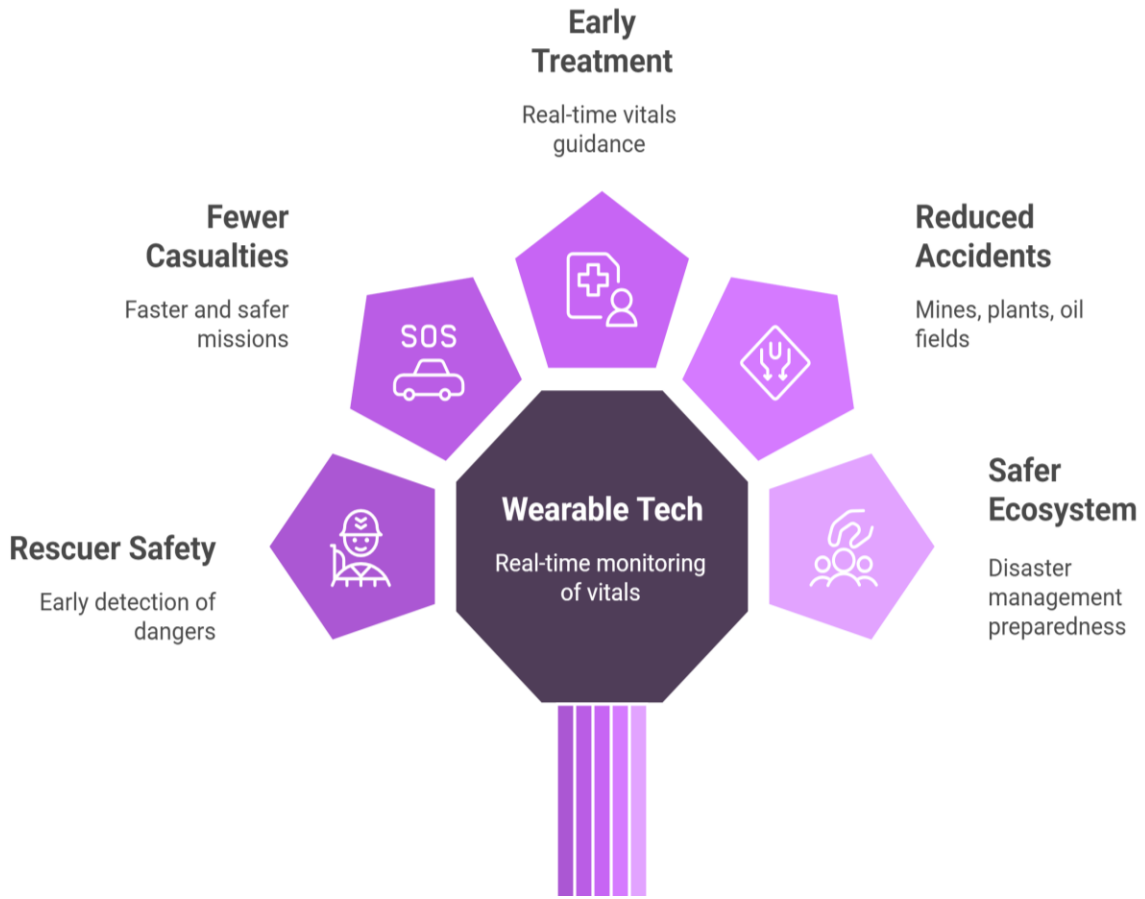
- Bluetooth/LoRa signals can be unstable in disaster zones.
- Balancing rugged protection with a lightweight design is challenging.
- Advanced sensors and rugged design may increase production costs.

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IMPACT AND BENEFITS

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IMPACT:



BENEFITS:



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RESEARCH AND REFERENCES

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IoT Smart Jacket for Health and Safety Monitoring || Aislyn Tech
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Smart Workwear
Grassi SPA https://www.grassi.it › en › abbigliamento-da-lavoro-intelligente

Smart Air Monitoring with IoT-based MQ-2, MQ-7, MQ-8, and MQ-135 Sensors using NodeMCU ESP32
(PDF) Smart Air Monitoring with IoT-based MQ-2, MQ-7, MQ-8, and MQ-135 Sensors using NodeMCU ESP32

Rescue Wear the Cutting Edge Wearable Technology for Emergency Operations
(PDF) Rescue Wear the Cutting Edge Wearable Technology for Emergency Operations

Wearable Devices for Rescue Operation
(PDF) Wearable Devices for Rescue Operation: A Review